

[521] (CONTINUED)

On Dr Wiener's Model of a Cubic Surface with 27 Real Lines.

p. 367

The model is a solid figure bounded by portions of the faces of the cube, and by a portion of the cubic surface, being a surface with three apertures, the orbiculation of which is not easily explained.

To determine the numerators I measured, on the faces of the cube, the coordinates of the two extremities of each of the twelve lines; these were measured in tenths of an inch (taking account of the half division, or twentieth of an inch); and the resulting numbers divided by 16 to reduce them to the belère-mentioned unit of $1 \cdot 6$ inches. These reduced values are shown in the table: knowing then the coordinates of two points on each line, the equations of the several lines became calculable; the true theoretical form of these results—viz. the form which, but for errors of the model, or of the measurement, they would have assumed—is

$$\begin{aligned}
 b_1 : \quad & x = R_2z + D, \quad y = R_2z + D, \\
 b'_1 : \quad & y = 0, \\
 b_2 : \quad & x = R_1(z + \beta_1), \quad y = R_1(z + \beta_1), \\
 & i = 1, \\
 b_3 : \quad & x = R_2(z + \beta_2), \quad y = R_2(z + \beta_2), \\
 b_4 : \quad & x = R_3(z + \beta_3), \quad y = R_3(z + \beta_3), \\
 b_5 : \quad & x = R_4(z + \beta_4), \quad y = R_4(z + \beta_4), \\
 & i = 1, \\
 a_1 : \quad & x = 0, \quad y = 0, \\
 a_2 : \quad & x = A_1z + C_1, \quad y = A_1(z - 1), \\
 a_3 : \quad & x = A_2(z + 1), \quad y = A_2(z - 1), \\
 a_4 : \quad & x = A_3(z + 1), \quad y = A_3(z - 1), \\
 a_5 : \quad & x = A_4(z + 1), \quad y = A_4(z - 1) + C_4,
 \end{aligned}$$

but in consequence of such errors, the results are not accurately of the form in question.

The faces of the cube being as in the diagram :

[Figure: unit cube with face labels — diagram omitted]

the Table is

p. 368

[Table: equations substituted from the measurements of the model, 17 columns of numerical values for $ABCD$, $EFGH$, $AEFF$, $HCFG$, $CDGH$, $ABHD$. — table omitted]

I hence calculate the intersections: considering any two lines which ought to intersect, then projecting on the principal plane and calculating x , y , the coordinates

p. 369

of the point of intersection of the two projections, these values of x, y substituted in the equations should give the same value of z ; but if the lines do not actually intersect, then the values of z will be different.

[Table: 7 column array b_1, b_2, b_3, b_4, b_5 with numerical intersection values — table omitted]

Starting from the assumed equations of b_1, b_2, b_3, b_4, b_5 , and calculating by the theory the remaining lines, the equations of the b -lines (those of b_1 being calculated) are

$$\begin{aligned} b_1 : \quad & y = 1284x + 154, \\ b_2 : \quad & x = 0, \quad z = -1, \\ b_3 : \quad & x = 1352(y + 510), \\ & y = 1009(z + 510), \\ b_4 : \quad & x = -731(z + 422), \\ & y = -500(z + 422), \\ b_5 : \quad & y = 0, \quad z = 1, \\ b_6 : \quad & x = 1131(z + 624), \\ & y = 1133(z + 624). \end{aligned}$$

p. 370

and the equations of the a -lines (those of all but a_1 being calculated) are

$$\begin{aligned}a_1 : \quad x &= 0, \quad y = 0, \\a_2 : \quad x &= -732z - 296, \\&\quad y = -820(z - 1), \\a_3 : \quad x &= -987(z + 1), \\&\quad y = -477(z - 1), \\a_4 : \quad x &= -2.506(z + 1), \\&\quad y = -541(z - 1), \\a_5 : \quad x &= -974(z + 1), \\&\quad y = -967z - 384, \\a_6 : \quad x &= -179(z + 1), \\&\quad y = -071(z - 1).\end{aligned}$$

and thence for the points of intersection the coordinates are

*[Table: 7 column array with $b_1, b_2, b_3, b_4, b_5, b_6$ headings and intersection-point coordinate values
— table omitted]*

p. 371

II.

I have in a paper "On the Orbi-tikere of a cubic surface," *Quart. Math. Journal*, t. x. (1870), pp. 58–71, [459], obtained analytical expressions for the twelve lines of a double-sixer, and also obtained numerical values, which however (as there remarked) did not come out convenient ones for the construction of a figure. A different mode of treatment since occurred to me, by means of the following equations of the cubic surface

$$\left(\frac{x}{a} - \frac{y}{c}\right) \left(\frac{x}{a} - \frac{y}{a}\right) \frac{z}{b} \left(\frac{x}{b} + \frac{y}{c} - \frac{z}{b}\right) \left(\frac{x}{b} - \frac{y}{a} - \frac{z}{b}\right) = 0,$$

which on self appear is a very convenient one for the purpose. We in fact obtain at once eight lines of the double-sixer; viz. those are

1. $x = 0, \quad w = 0,$	I'. $x = 0, \quad y = 0,$
2. $y = 0, \quad z = 0,$	II'. $z = 0, \quad w = 0,$
3. $\frac{x}{a} - \frac{y}{c} = 0, \frac{w}{b} - \frac{y}{a} = 0,$	III'. $\frac{x}{a} - \frac{y}{a} = 0, \frac{y}{c} - \frac{z}{b} = 0,$
4. $\frac{x}{a} + \frac{y}{c} = 0, \frac{w}{b} - \frac{y}{a} = 0,$	IV'. $\frac{x}{a} - \frac{y}{a} = 0, \frac{y}{c} + \frac{z}{b} = 0,$

and also five lines not belonging to the double-sixer, viz.

12. $x = 0,$	$\left(-\frac{x}{a} + \frac{y}{c} + \frac{z}{b}\right) \frac{1}{abc} + \frac{w}{B} \left(-\frac{x}{a} + \frac{y}{c} - \frac{z}{b}\right) \frac{1}{ABC} = 0,$
23. $y = 0,$	$\left(\frac{x}{a} + \frac{y}{c} - \frac{z}{b}\right) \frac{1}{abc} + \frac{w}{B} \left(\frac{x}{a} - \frac{y}{c} - \frac{z}{b}\right) \frac{1}{abc} = 0,$
34. $z = 0,$	$\left(\frac{x}{a} - \frac{y}{c} - \frac{z}{b}\right) \frac{1}{abc} + \frac{w}{B} \left(\frac{x}{a} + \frac{y}{c} - \frac{z}{b}\right) \frac{1}{abc} = 0,$
41. $w = 0,$	$\left(\frac{x}{a} - \frac{y}{c} + \frac{z}{b}\right) \frac{1}{abc} + \frac{w}{B} \left(\frac{x}{a} + \frac{y}{c} + \frac{z}{b}\right) \frac{1}{abc} = 0,$
5 & 2. $\frac{x}{a} + \frac{y}{c} = 0,$	$\frac{z}{b} + \frac{w}{B} = 0.$

The remaining lines of the double-sixer are then easily determined; viz. the lines 3, 5, 6, and 12 are met by the line Γ , and by a second line Γ' ; this, as a line meeting 3, 5, 6, will be given by equations of the form

$$x = \frac{y}{a}(a+b), \quad x = \frac{y}{c}(a-c),$$

and observing that these equations, writing therein $z = 0$, give

$$\frac{x}{a} - \frac{y}{c} = 0, \quad \frac{y}{c} - \frac{z}{b} = 0,$$

p. 372

the condition of intersection with the line 12 gives

$$\phi = \frac{a+b}{c-1} \cdot \frac{c-2}{a-1},$$

which is the value of ϕ in the foregoing equations; and so these we may join the resulting equation

$$pqr(aB + cB) = abqB(qB + rB),$$

Proceeding in like manner then for the lines 1, 2, 4, 5, the equations for the remaining four lines of the double-sixer are; those are

2. $\phi = \frac{a-bc}{B-bc},$ $x = \frac{y}{a}(a + \frac{B}{a}),$ $z = \frac{a}{b}(a + \frac{B}{a}),$ $pqr(aB + cB) = abB(rB - dB),$	1'. $\phi = \frac{c-ba}{B-ba},$ $x = \frac{y}{a}(a + \frac{c}{B}),$ $z = \frac{a}{b}(a + \frac{c}{B}),$ $pqr(aB + cB) = abB(pB - qB),$
4. $\phi = \frac{a-bc}{B-bc},$ $\phi\left(\frac{x}{a} - \frac{w}{b}\right) = y,$ $\phi\left(\frac{x}{a} - \frac{w}{B}\right) = \frac{a}{c},$ $pqr(aB - cB) = qBa(BqB - dB),$	3'. $\phi = \frac{c-2a}{B-2a},$ $\phi\left(\frac{x}{a} - \frac{w}{b}\right) = -\frac{y}{c},$ $\phi\left(\frac{x}{a} - \frac{w}{B}\right) = -\frac{a}{c},$ $rqr(aB - cB) = abr(rB - 2dB).$

It may be added that —
in plane $z = 0$, the

$$\begin{aligned} &\text{intersection of } \Gamma \text{ line on line 2 : } y = (cB - cD)cy : sqrB - aR, \\ &,, \quad 2 \quad ,, : y : x = -cyB(cD - Db)sw : sqrB - aR, \end{aligned}$$

and that the line joining these intersections is the line 12.

In plane $y = 0$,

$$\begin{aligned} &\text{intersection of } 1' \text{ line on line } x : y : z = pqrB - aR : qprB - aR, \\ &,, \quad 1 \quad ,, : x : z = wrB(aR - 2Bdy)sw : qrpB - aR, \end{aligned}$$

and that the line joining these intersections is the line 23.

p. 373

In plane $z = 0$,

$$\begin{aligned} &\text{intersection of } 3' \text{ line on line } x : y = (yB - rj)sw : sqrB - cyB; \\ &,, \quad 4 \quad ,, : x : w = -(Dy - By)sw : sqrB - cyB; \end{aligned}$$

and that the line joining these intersections is the line 34.

And in plane $w = 0$,

$$\begin{aligned} &\text{intersection of 4 line on line } x : y : z = wpyB - cyB : ywpyB - cyB; \\ &,, \quad 1' \quad ,, : x : y = wpyB - cyB : -ywpyB - cyB; \end{aligned}$$

and that the line joining these intersections is the line 41.

The equations of the remaining ten lines of the surface may be obtained without difficulty, and also the thirty further triple planes; but I do not stop to effect this. The construction of a model it is sufficient to determine the three points on each edge, and the two points, on the planes of $ABCD$, viz. the points a_0, b_0, c_0, d_0 , in which the three sides of the triangle ABC meet by triple planes, and the points ABC, a_0BCD (for the construction of the remaining ten lines of the surface by another method, viz. employing the points $A, B, C, D, a_0, b_0, c_0, d_0$). In fact, knowing each of $a, b, c, d, \alpha, \beta, \gamma$ on the edges of the cube, we know the values of ξ, η, ζ for the points on the edge BC ; the points on the edge BC are obtained from the corresponding points on AD by the lines 4 and 1. For the construction of a model it is sufficient to determine the three points on each edge, and the two points, on the planes of $ABCD$, viz. the points a_0, b_0, c_0, d_0 , in which the three sides of the triangle ABC meet by triple planes, and the points $abAB, bdAD, caCD$ on the planes of ABC . Then for ab we have

$$\frac{x}{a} + \frac{y}{c} = 0, \quad \frac{y}{c} - \frac{z}{b} = 0, \quad w = 0,$$

viz. the $ABCD$ -coords mementos of the orbi parametrised the lines 1, 2, 3, 4. But if any sphere of an cube and several distances mentioned in the orbi- clear in their intersection with the line 4'; and it is probable that a slightly different value of ϕ would be better.

[Figure: triangle with three points — diagram omitted]

p. 374

The results just obtained may be exhibited in a compendious form as follows:

[Table: 5 column array with column headings $\alpha, \beta, \gamma, \delta, \theta$ listing intersections for various points 1, 2, 3, 4 etc with AB, AD, ACD , etc. — table omitted]

p. 375

Γ' and Γ meet BCD on line: $\left(-\frac{x}{a} + \frac{y}{c}\right) \frac{1}{abc} = \left(-\frac{x}{a} + \frac{y}{c} - \frac{z}{b}\right) \frac{1}{ABC} = 0$,

2 and $3'$ „ CDA „ $\left(\frac{x}{a} - \frac{y}{c}\right) \frac{1}{abc} = \left(\frac{x}{a} - \frac{y}{c} - \frac{z}{b}\right) \frac{1}{ABC} = 0$,

3 and $4'$ „ DAB „ $\left(\frac{x}{a} - \frac{y}{c}\right) \frac{1}{abc} = \left(\frac{x}{a} - \frac{y}{c} - \frac{z}{b}\right) \frac{1}{ABC} = 0$,

4 and $1'$ „ ABC „ $\left(\frac{x}{a} - \frac{y}{c}\right) \frac{1}{abc} = \left(\frac{x}{a} - \frac{y}{c} - \frac{z}{b}\right) \frac{1}{ABC} = 0$,

For calculating the numerical values from the foregoing assumed data, say

[Table: column headings $\alpha, \beta, \gamma, \delta$ with numerical computation entries — table omitted]

p. 376

which had two equations now as a verification.

The outside numerical values are given in the manner most convenient for the construction of a drawing; viz. when the coordinates refer to a point on an edge of the tetrahedron, or any on the side of an equilateral triangle, the lines being the length of this edge (or side) to be = 100, the numerical values are fixed so that the answers of the points (regarded as = 0, 20, 40, ..., 100) are situated in the table similarly to that of the extension of the edge or side: but when the three coordinates belong to a point in the face of the tetrahedron, or any in the plane of an equilateral triangle, then the sum of the coordinates is made = $80 \cdot 6$, and the three coordinates these denote the perpendicular distances from the sides of the triangle.

p. 377

III.

It is possible to find on a cubic curve a double-sixer of points $1, 2, 3, 4, 5, 6$ and $1', 2', 3', 4', 5', 6'$ such that any six points such as $1, 2, 3, 4, 5, 6$ lie in a conic. In fact considering a cubic surface having upon it the double-sixer of lines $1, 2, 3, 4, 5, 6$ and $1', 2', 3', 4', 5', 6'$, the section by any plane is a cubic curve meeting the lines, say in the points $1, 2, 3, 4, 5, 6, 1', 2', 3', 4', 5', 6'$; each of the six lines $1, 2$ meets each of the six lines $1', 2', 3', 4', 5', 6'$, and consequently the six lines lie in a quartic surface; therefore the points $1, 2, 3, 4, 5, 6'$ lie in a conic; and so in like order; cases; the number of the conics is of course $= 6$.

The cubic curve may be a given curve, and six of the points upon it (not being points on a conic) may be taken to be, given; for instance the points $1, 2, 3, 1', 2', 3'$. For take through the points $1, 2$ respectively any two lines $1, 2$; through $1', 2', 3'$ respectively the lines $1', 2', 3'$ each meeting each of the lines $1, 2$; and through 1 a line meeting each of the lines $1, 2, 3$, it is easy to see that a cubic surface may be drawn through the cubic curve and the lines $1, 2, 3, 1', 2', 3'$. In the passage through the cubic curve requires only 9 conditions; the surface then passes through the point 2 and so make it pass through the line 2 requires 3 conditions; similarly the surface passes through the point 3 , and to make it pass through the line 3 requires 3 conditions. The surface now passes through 1 and through the points of intersection of the line $1'$ with the lines $2, 3$; to make it pass through the line $1'$ requires 1 condition; similarly to make it pass through the lines $2', 3'$, 1 requires in each case 1 condition; or there are in all 19 conditions, so that the cubic surface is completely determined. Take now through the points $1, 2, 3, 4, 5'$, a conic meeting the cubic in the points $4', 5'$, then through the lines $1, 2, 3, 4', 5'$ we have a quintic surface passing through the

p. 378

conic, and therefore through 4'; hence through 6' we may draw a line 6' meeting each of the lines 1, 2, 3, 1', 2', and where the cubic surface passes through the point 6' and also through the intersections of the line 6 with the lines 1, 2, 3; it passes through the line 6'. We complete in this manner by determining the planes of the cubic the system of the twelve points, viz. each new point is given as the intersection of the cubic curve by a conic drawn through five points of the cubic curve. It is then shown as for the point 1' and the line 6' through it, that through each new point there can be drawn a line meeting by the same number and meeting each of the lines which it ought to meet; and hence lying on the cubic surface, but the twelve points are thus the intersections of the planes of the cubic curve by the twelve lines of the double-sixer, and it follows that the six points which ought to lie in a conic do every one where each conic has not been used in the place construction do actually lie in a conic.

I was anxious to construct such a double-sixer of points on a cubic curve; for this purpose I take the equation of the curve to be $y' = (1 - \frac{x}{a})(1 - \frac{x}{b})(1 - \frac{x}{c})$, or say for shortness, $y^2 = X$; where, to fix the ideas, a, b are supposed to be positive, c greater than b ; and r to be negative.

The cubic curve is thus a parabola symmetrical in regard to the axis of x , and consisting of a loop and infinite branch; say I select for them the points 1, 2, 3, 1', 4', 5'

[Figure: parabolic loop with axis through marked points — diagram omitted]

as shown in the figure, viz. the coordinates of these points are as stated in the Table, where α is the x coordinate, and $\sqrt{M} = a_0 \sqrt{(1 - \frac{x}{a})(1 - \frac{x}{b})(1 - \frac{x}{c})}$ and so in other cases, $\sqrt{14} = 374483$.

p. 379

The numerical values belong to the curve $y' = \left(1 - \frac{x}{1}\right) \left(1 - \frac{x}{2}\right) \left(1 + \frac{x}{3}\right)$, and to $a = 6$.

Starting with the points 1, 2, 3, 1', 4', 5' we have to find the remaining points 4', 5, 6, 2', 3', 6'.

Point 4' by means of the conic 12345'6', as follows.

The equation of the conic is

$$(x - b)(x - c) - bcy + xy = 0 \quad (1, 2, 3, 4, 1'),$$

and making this pass through the point 1 ($x = a, y = \sqrt{M}$) we find

$$m_0 = (a - b)(a - c) + \sqrt{M} = 0 \quad (1).$$

Hence taking the coordinates of 4' to be, $a_0, \sqrt{M_0}$, we have

$$(a_0 - b)(a_0 - c) - bca_0 + \sqrt{M_0} = 0 \quad (4'),$$

and thence

$$\frac{\sqrt{M_0}}{\sqrt{M}} \cdot \frac{(a_0 - b)(a_0 - c) - bca_0}{(a - b)(a - c) - bca} = 1.$$

p. 380

or say

$$F(\beta - 1) + G\sqrt{M} = \frac{a - q}{B - q} \cdot \frac{1}{a},$$

that is,

$$(\beta - 1)abcF + ac\sqrt{M} = \frac{1}{ab}(\beta - qa^2a^2 + a)$$

viz. that is,

$$(\beta - 1)(F - \theta) + (a - bc)\frac{1}{\sqrt{M}}(F^2 + ac) = 0$$

or, rationalising and throwing out the factor $\theta - b$, this is

$$(\beta - 1)(F - q) + b(\beta - bc)(a - q)\frac{ac}{\sqrt{M}}(F^2 + ac) = 0,$$

which is a cubic equation satisfied by $\theta = a$ and $\theta = a_0$, so that throwing out the factors $\theta = a$, $\theta = a_0$, we have for θ a linear equation.

Putting for shortness

$$\begin{aligned} A &= (a - a)F + (a - b)(a - c), \\ B &= (a - b)F + (b - c)(a - a), \\ C &= (a - c)F + (a - c)(a - b), \end{aligned}$$

the value of θ may be expressed in the forms

$$\theta = a + \frac{aA}{BC}(F + a), \quad \theta = b + \frac{aA}{BC}(F + b), \quad \theta = c + \frac{aA}{BC}(F + c).$$

We have moreover

$$F - q = \frac{a(a - 1)(a - b)(a - c)}{BC}, \quad F - a_0 = -\frac{ac - A}{B}.$$

equations which express F in terms of a only; also

$$\theta - F = -\frac{a(a - 1)(a - b)(a - c)B}{C^2},$$

and then

$$\sqrt{M_0} = \sqrt{M} \cdot \frac{B^2 + ac}{C^2},$$

whence

$$\sqrt{M_0} = \pm\sqrt{M}(b - c) - \frac{aB}{C^2},$$

so that θ , $\sqrt{M_0}$ can be determined.

p. 381

Point 5 by means of the conic 1234'5'6'.

The conic is

$$Fx + Gy + H = \frac{1 - y'}{x}, \quad (1, 2, 3)$$

where

$$\begin{aligned} Fx + H &= 1, \\ Fa_0 + G\sqrt{M} + H &= \frac{1 - M_0}{a_0}, \quad (4'), \\ Fa_0 - G\sqrt{M_0} + H &= -\frac{M_0}{a_0}, \quad (5'). \end{aligned}$$

Everything is the same as for the point 4 except that b, c are interchanged; hence writing Q instead of F , and using A, B, C to denote as before, we have

$$\begin{aligned} abcF &= a + b + Q, \\ \sqrt{M}abcG &= (a - c)(1 - a + bQ), \\ abcH &= ab - aa + bQ - aB, \end{aligned}$$

and

$$\begin{aligned} \phi &= \frac{C^2 - a}{BC}, \\ \phi &= b - \frac{a(b - a)(b - 1)b - c}{B} - \frac{aa(b - c)(b - 1)(b - a)}{BC}, \\ Q &= b + \frac{B^2 + ac}{BC}, \\ Q - a &= \frac{B(b - c)(b - 1)(b - a)}{BC}, \\ Q - a_0 &= \frac{B(a - c)(b - 1)(b - a)}{BC}, \end{aligned}$$

and

$$\sqrt{M_0} = \pm\sqrt{M}(b - c) - \frac{aQ}{BC^2}.$$

which determine $\phi, \sqrt{M_0}$.

Point 5' by means of the conic 12345'6'6'.

The conic is

$$Fx + Gy + H = \frac{(a - b)(a - x)}{y}. \quad (a, 1)$$

p. 382

and we have

$$\begin{aligned} G + H &= a, \\ Fa_0 + G\sqrt{M} + H &= \frac{(a-b)(a-a_0)}{\sqrt{M}}, \quad (1) \\ Fa_0 - G\sqrt{M} + H &= -\frac{(a-b)(a-a_0)}{\sqrt{M}}, \quad (1) \end{aligned}$$

Eliminating F , we have

$$G(aa_0\sqrt{M} + aa\sqrt{M_0}) = aa_0 + \frac{a_0(a-b)(a-a_0)}{\sqrt{M}}\sqrt{M} + \frac{a(a-b)(a-a_0)}{\sqrt{M}}\sqrt{M_0},$$

which is easily reduced to the form

$$G \frac{\theta a_0 - \theta aa_0 + a aa_0}{(aa - a)\sqrt{M}} + H(\theta - a) = (a + a) + 2a\sqrt{M_0},$$

and thence

$$G(aA + 2ac(\theta - a)) - \frac{\theta a_0 - \theta aa_0 + a aa_0}{\sqrt{M}}\sqrt{M_0} + abc + 2bc(a - a_0)\sqrt{M_0}$$

and combining herewith $G + H = b$, we have

$$H = \frac{2bc(a - a_0) - a + ba_0 + ac}{(a - a) - aA}\sqrt{M_0} + \frac{aA(b - a) + 2abc(aa_0 - a_0 + aa)}{aA(a - a) - aA},$$

and we have then

$$G = b - H;$$

that is,

$$F(a - a_0) + G\sqrt{M} + 2H = 0,$$

that is, what is the same thing,

$$F(aa_0(a - a_0) - 2bH + \frac{A}{C} + \frac{2aA}{C^2}\sqrt{M_0} + 2H(a - a_0) = 0.$$

We have therefore

$$\begin{aligned} Fa + H &= \frac{a}{2} \left(G - \frac{Aa(aaa_0 - aA)}{a - aa_0} \right) \\ &= \frac{1}{2} \left(b - G + \frac{Aa(aA + a)}{a - aA} \right), \end{aligned}$$

that is,

$$(F + H)x = \frac{(a-b)(a-x)}{xba}(Hx + 2Gx).$$

p. 383

or

$$aba_0(a - a_0)F + Hx + 2Gx = (a - b)(a + a_0)(Hx + 2Gx) = 0.$$

Developing and throwing out the factor $a - b$, we have

$$\begin{aligned} G^2a^2 + [(Baac) - (b + c)Gaaba^2] + [a^2H^2 - 2a(b + c)GH + a(ba^2 + a^2H^2)] \\ + [-(b + c)a^2H^2 + bacGH + a(BaFH)] = 0; \end{aligned}$$

This must be satisfied by $a = a$, $a_0 = a$; hence the left hand must be $= a^2(b - a)(a - aa_0)$, or equating the constant terms we have

$$G^2aaa_0 = aA(1 - 2aacB(b - a)) - a_0(B - 2baFH + acH^2),$$

which gives a^2 ; and we then have

$$\sqrt{M_0} = \frac{a - a_0}{aB - aA} (Faa + H),$$

but I have not attempted the further reduction of these expressions.

The numerical values for the example are

$$3F = \frac{-180 + 62\sqrt{14}}{5 + 21\sqrt{14}}, \quad G = \frac{-10 + 62\sqrt{14}}{5 + 21\sqrt{14}}, \quad H = \frac{-104\sqrt{14}}{1 + \sqrt{14}}$$

whence a_0 is in the Table.

Point 2' by means of the conic 1 34'5'4'1'.

The equation of the conic is

$$Fx + Gy + H = \frac{a - b}{y}, \quad (a', 1)$$

where

$$\begin{aligned} -G + H &= ba_0, \\ Fa_0 + G\sqrt{M_0} + H &= \frac{(b - a)(a - a_0)}{\sqrt{M_0}}, \quad (1) \\ Fa_0 - G\sqrt{M_0} + H &= \frac{(b - a)(a - a_0)}{-\sqrt{M_0}}, \quad (1), \end{aligned}$$

which are the same as for point 5', but only we recover the signs of F , H , and $\sqrt{M}$, $\sqrt{M_0}$.

p. 384

Hence the formulae are

$$H = \frac{2bc(a - a_0) - a + ba + ac}{aA + 2ac(a - a) - (a)} \sqrt{M_0} + \frac{aA(b - a)}{aA},$$

$$G = ba + H,$$

$$F(a - a_0) + G\sqrt{M_0} - 2bc(F + H)a_0,$$

which gives a ; and then

$$Gaa_0\sqrt{M_0}\sqrt{M_0} = -3aac(Fa + H)\sqrt{M_0} + H(Fa + H),$$

which are also reduced.

The numerical values are

$$3F = \frac{140 + 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad G = \frac{-10 - 62\sqrt{14}}{5 - 21\sqrt{14}}, \quad H = \frac{-104\sqrt{14}}{3 - 21\sqrt{14}},$$

whence θ is in the Table.

[522]

Note on the Theory of Invariants.

[From the *Mathematische Annalen*, vol. III. (1871), pp. 268–271.]

p. 385

IF two binary quantics $(a, \dots)(x, y)^n$, $(a', \dots)(x', y')^n$ are linearly transformable the one into the other, and if for the first of these P, Q are any two invariants whatever of the same degree, and P', Q' are the like invariants for the second of them, then we have

$$P : Q = P' : Q'.$$

viz. what is the same thing, the absolute invariants have the same values for the two functions respectively; and the entire system of these equations constitutes only a $(n - 3)$ -fold relation between the two sets of coefficients. For the converse theorem, viz. that if the entire system of equations is satisfied, the two functions are linearly transformable, see Clebsch, “*Theorie der binären algebraischen Formen*,” § 89.

For instance, considering the two binary quartics

$$(a, b, c, d, e)(x, y)^4, \quad \text{and} \quad (a', b', c', d', e')(x', y')^4,$$

say with the invariants

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3, \end{aligned}$$

and the like for the second of them; the conditions of equivalence are

$$I : J = I' : J',$$

or, what is the same thing, $I^3 J^{-2} = I'^3 J'^{-2}$, viz. this is a single relation between the two sets of coefficients (a, b, c, d, e) and (a', b', c', d', e') , so that any 4 of the parameters (a', b', c', d', e') may be considered as homogeneous functions of the elements (a, b, c, d, e) of the form

$$P : Q = P' : Q'.$$

p. 386

To deduce this result from the theory of the cubic function, I observe that denoting by A, B, C, Δ the values of the quadrinvariant, the sextinvariant, and the discriminant, as given in Salmon's *Higher Algebra*, Ed. 2, pp. 202–211, then in the particular case $a = 0, b = 0$, we have

$$\begin{aligned} A = 10B = c^2, \quad C = -8d^2, \quad \Delta = 0, \\ + 13ce, \quad -3cde \quad + 36cd^2e \\ + d^2, \quad +9d^2, \quad -27d^4e^2 \\ + e^2, \end{aligned}$$

and hence forming the new invariants

$$\begin{aligned} B - 100B = AI, \\ \Gamma = 1000C - 1200AB + ??A^3. \end{aligned}$$

the values of these in the same particular case $a = 0, b = 0$ are

$$\begin{aligned} B - 25ce(Icf - bg) = I = 200c^3e + 9ce - 12def + cd, \\ -200cd^4e + 8000d^2(3ce + d^2), \end{aligned}$$

Taking now A, B, C, Δ as the invariants of the cubic, one of the conditions for the transformation is $B^3 : C^2 = B'^3 : C'^2$.

In the particular case $a = 0, b = 0$ and $a' = 0, b' = 0$, the invariants vanish and the equation is satisfied identically. But if we assume in the first instance only $a = 0, b = 0$, but not $a' = 0, b' = 0$, then the terms which contain the common factors c' and c' respectively; and throwing these out, and then writing $c = 0, d = 0$, we obtain previously found in a different manner.

It will be observed that the condition is of the original form $P : Q = P' : Q'$, but with the difference that P, Q and the corresponding functions P', Q' are not invariants. In presenting the foregoing property these functions may however be called “imperfect invariants”; it being understood that an imperfect invariant is not an invariant, and is not in any case included in the term “invariant” used without qualification.

And we may now establish the general theory as follows: Consider the similarly constituted special forms $(a, \dots)(x, y, z, \dots)^p$ and $(a', \dots)(x', y', z', \dots)^p$; to be in these the coefficients $(a, \dots)$ may be regarded as homogeneous functions of the elements $(z, \dots)$, which are either algebraical, or homogeneously connected together in any manner; and then the coefficients $(a', \dots)$ will be the like functions of the elements $(x', y', \dots)$ which are either independent or (as the case may be) homogeneously connected in the like manner.

The entire series of functions $P, Q, \dots$ of $(a, \dots)$, which are such that, if for any of the same degree, and P', Q' being the like functions of $(a', \dots)$, the relations $P : Q = P' : Q'$ are equivalent to a $(n - 3)$ -fold relation between the linearly transformable function $(a, \dots)(x, y, z, \dots)^p$, may, in a sort of like manner,

$$P : Q = P' : Q'.$$

p. 387

may be called the “perfect and imperfect invariants” of $(a, \dots)(x, y, z, \dots)^p$; and the relation in question be briefly referred to by the expression that the perfect and imperfect invariants are proportional.

We have then the theorem that if the two functions $(a, \dots)(x, y, z, \dots)^p$ and $(a', \dots)(x', y', z', \dots)^p$ are linearly transformable the one into the other, the two functions have their perfect and imperfect invariants proportional; and *conversely* the theorem, that two functions which have their perfect and imperfect invariants proportional, are linearly transformable the one into the other.

There is from a wide field of inquiry in regard to the imperfect invariants, even of a binary function, but still more so as to those of a ternary or quaternary function representing a curve or surface possessed of singularities.

We have in what precedes the explanation of an error into which I fell in my paper “On the transformation of plane curves,” *Proc. Lond. Math. Soc.*, vol. I. No. 3, Oct. 1865, [384], see Arts. Nos. 27–30. Considering a given curve of deficiency D and, by means of a system of $D - 5$ points chosen at pleasure on the curve, transforming this into a curve of the order $D + 1$ with deficiency D ; then for any two of the transformed curves (that is, two curves obtained by means of different systems of the $D - 5$ points) I showed that these had the same absolute invariants, common factors σ' and σ'' respectively; and throwing these out, and then writing $c = 0$, $d = 0$, we have $cc^2dc^2 = c'c'^2dc'^2$, viz. H vanish.

I remark that if the two transformed curves are linearly transformable the one into the other, then it cannot in the present case be true.

We have a class of surfaces of negative deficiency; viz. any rational transformation of a cone for which the planes section has a given (positive) deficiency produces such a surface; and I think it may be answered conversely that a surface of negative deficiency is always the rational transformation of a cone for which the deficiency is equal to that of the surface taken with the reverse sign. As an instance, of course implies that the line joining the two corresponding (rational) points (this of course implies that the line joining the two corresponding points of the surface). We may call these “perfect and imperfect invariants”; it being understood that an “imperfect invariant” has all the properties of an invariant except that it does not satisfy the conditions of the linear transformation; and now in regard to the existence in the case of the hyperboloid is readily perceived.

Cambridge, 4 August, 1870.

[523]

On the Transformation of Unicursal Surfaces.

[From the *Mathematische Annalen*, vol. III. (1871), pp. 469–474.]

p. 388

I CONSIDER the question of the transformation (Abbildung auf einer Ebene) of unicursal surfaces. Taking (x, y, z, w) for the coordinates of a point on the surface, (x', y', z') for those of the corresponding point on the plane; then if X', Y', Z', W' denote each of them a function $(x', y', z')^n$, the equations of transformation are

$$x : y : z : w = X' : Y' : Z' : W';$$

and assuming that each of the curves

$$X' = 0, \quad Y' = 0, \quad Z' = 0, \quad W' = 0$$

(or, what is the same thing, the general curve

$$\alpha X' + \beta Y' + \gamma Z' + \delta W' = 0)$$

passes once through each of α points, twice through each of β points, $\dots$, r times through each of ν points, the curves the successions Γ write σ , instead of $\alpha, \beta, \dots$), and writing also

$$n = n^2 - \Sigma r^2 \alpha,$$

$$0 = 4n^2(n - 3) - 3 - n - \Sigma r(r + 1)\alpha,$$

(where $0 = 1$ as positive except in the case of special relations between the positions of the fixed points $a, n_1, \dots, n_n$, which equations give

$$-n = 3n - 6 - 28n - 2\Sigma \alpha,$$

then the order of the surface is n , and the order of the nodal curve is $0 = 2n - \Sigma r - 2\Sigma w$; I assume that the nodal curve has ρ apparent double points and ν actual triple points, but no stationary points, so that q being the class, we have $q = D - k - 2\delta - 3\nu$; and I endeavour to find from these numbers $\alpha, n_1, \dots, n_n$.

p. 389

For this purpose, imagine through the nodal curve a surface of the order k , which therefore meets the surface besides in a curve of the order $nk - 25$; this curve Γ and the cubic curve of the nodal curve, or simply the "residue." The projection (Abbildung) of the complete intersection $5n$ is a curve of the order kn' passing through each of the projections of the residue. But as shown by Dr Clebsch the projection of the nodal curve is of the order $(n-4)n' + 3$, and it passes $(n-4)r + 1$ times through each of the points a_1 ; hence the projection of the residue is of the order $(k-n+4)n' - 3$, and it passes $(k-n+4)r - 1$ times through each of the points a_1 . I assume that the projection of the residue is the *general* curve which satisfies the foregoing conditions, viz. that the residue, and its projection as defined by the foregoing conditions, *depend each of them on the same number of constants*. The necessity for this I confess by no means obvious: but take as an illustration Steiner's quartic surface as transformed by the equations $x : y : z : w = x^2 : y^2 : z^2 : (x' + y' + z')$; the nodal curve consists of three lines meeting in a point, the quadric residue is the remaining intersection of the surface by a quadric cone passing through the three lines; and the projection thereof is a line; the quadric cone, and therefore the conic, each depend upon 2 constants; and the line which is the projection of the conic depends also upon 2 constants; that the lines which is the projection of the conic depends the same number (2) of constants : at all events I make the assumption provisionally.

Now in the projection of the residue, we have twice the number of constants

$$= [(k-n+4)n' - 3](k-n+4)n' - (k-n+4)r\alpha_1,$$

viz. this is

$$= (k-n+4)^2 n^2 + 4(k-n+4) - 3n(k-n+4)$$

or, what is the same thing, is

$$= (k-n+4)^2 n + (k-n+4)(n-4) + 2(k-n+4),$$

viz. reducing by means of the relation $-n = 4n - 6 - 28 + 2(n-4)$, the number in question is

$$= k^2 n + k(-n^2 + 4n - 26) + 2(n-4)k.$$

Now k being $=$ or $> n - 3$, the curve of intersection of a given surface as by a surface k depends on

$$\frac{1}{2}(k+1)(k+2)(k+3) - \frac{1}{2}(k-n+1)(k-n+2)(k-n+3) - 1$$

constants; and making the surface k to pass through the nodal curve, we have to subtract herefrom $(k+1)b - \frac{1}{2}b - 2k$; that is, for the residue, twice the number of constants is

$$= \frac{1}{2}(k+1)(k+2)(k+3) - \frac{1}{2}(k-n+1)(k-n+2)(k-n+3) - 2 - 2(k+1)b + \delta + 4t,$$

viz. this is

$$= k^2 n + k(-n^2 + 4n - 2b) + \frac{1}{4}(n-2)(n-3)(n-4) - 2b - 25 + q + 2(n-4).$$

p. 390

Hence comparing the two expressions in question we have

$$2(n-4) = (n-4)n' - \frac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q,$$

that is,

$$0 = (n-4)\{n' - \frac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q\} = q,$$

or, as I prefer to write it,

$$0 = (n-4)\{n' - \frac{1}{4}(n-2)(n-3)(n-4) + 2b + 25 - q\} = 0;$$

which agrees with a more general formula in my "Memoir on the theory of Reciprocal Surfaces," *Phil. Trans.* vol. CLI. (1869), [167], see p. 327, [Coll. Math. Papers, vol. p. 316]. I consider any two residues, a b -fold residue and a b' -fold residue, the intersection of these two surfaces and surface be equal to that of the two projections. Now the projection being (as above) twice the

order $(k-n+4)n' - 3$ passing through $(k-n+4)r - 1$ times through each of the points a_1 ,

the number of the intersections is

$$= (k-n+4)(k'-n+4)n^2 - 3(k-n+4) - 3(k'-n+4) + 9 - \Sigma\{(k-n+4)r-1\}\{(k'-n+4)r-1\}$$

viz. this is

$$= (k-n+4)(k'-n+4)n + (k-n+4)(n-4) + (k'-n+4)(n-4) - \Sigma(2k-n)(k'-n+4)$$

viz. substituting for θ its value $= n^2 - 4n + 2\Sigma r - 2 + 2n$, and reducing, this number is

$$= 4k - 2k - k^2 - 1 + bn'k - 6n - 2bk - (5n^2 - 25n + 12) - 2(n-4).$$

But the surfaces k, k' , having in common the curve b which is a nodal curve on n , besides intersect in

$$bk - (b+1)k' + 25 + 27 - 4t + 2q + p$$

the number of points (*Salmon's Geometry of Three Dimensions, 2nd Ed., p. 283, except that in the formula as there* $bk - 1 + (b+1)k' - 4t + 2q + p$ which is equal to the number of intersections of the two projections.¹

¹I remark that there a being a positive number, b is positive in each case I subtract herefrom $bk - 1 + (b+1)k' - 4t + 2q$, the residue of the term so for $bk - 1 + (b+1)k' - 4t + 2q$, and the rest of points are $bk - 1 + (b+1)k' - 4t + 2q$.

p. 391

But we have already found

$$2q + 4t + p = 4(n - 5) - \frac{1}{2}n^2 + 4n^2 - bn^2 + bn,$$

and we have therefore

$$t = (n - 5) - \frac{1}{2}n^2 + 4n^2 - bn^2 + bn + bn,$$

and

$$q = -2(n - 4) - \frac{1}{2}n^2 + 4n^2 - 5n + 4(5n - b).$$

I obtain these results in a different manner by investigating expressions for the deficiency (Geschlecht) of the nodal residue = 28 and for that of its projection.

First for the projection, the curve here is

$$\begin{aligned} 2D &= [(n - 4)n^2 - 3](n - 4)n^2 - 4 - \Sigma(n - 4)r^2 - 1)(n - 4)r, \\ &= 2(n - 4)(n - 5) + (n - 4) - 4 + n^2 - 4n^2 - 5n + 4(5n - b), \end{aligned}$$

where, as before, $\Sigma r^2 = n^2 - n$, $\Sigma r = -n$, and $\Sigma r^2 - \Sigma r = n - n$. To apply this to the deficiency D of the nodal residue, we have

$$q = -2(n - 4) - 1 = n^2 - 4n + 2\Sigma r - (3 - 25n - 2n),$$

or substituting for θ in this value, $= [(n - 4) - \frac{1}{4}(n - 1)(n - 2)(n - 3) + 2b + 25 - q]$, this is

$$= 2(n - 5) - (n^2 - 4n^2 + 4n - 4t + 2q)(n - 4),$$

Next as regards the residue, the number of its apparent double points is obtained in terms of h and t by the formula

$$bk - n(k - 1) = (b + 1)(b + 2) + 2\delta - 4t \quad (\text{Salmon, l. c., p. 284}),$$

except that the singularity t is not there taken account of; that is, the number of intersections of the two residues is

$$2\delta = bk^2 - (k - 1)bk + 2(b + 1)b + 2\delta - 4t - 4t.$$

or introducing a factor of 2 in the value of θ , $= (n - 4)\{(n - 5n^2 + 4n - 4) + bn^2 + bn$. viz. this is

$$2\delta = (n - 4)\{(n - 4n^2 + 4n - 4) - 24 + bn^2 + bn^2 - 4t,$$

viz. this is

$$2\delta = (n - 4)\{(n - 4n^2 + 4n - 4)\} - 24n + bn^2 + bn^2 + (n - 5n^2 + 4n - 4)\}.$$

the same result as before.

p. 392

The necessity for the term ω appears by the consideration that if we apply to the plane figure a Cremona-transformation, thus obtaining a new transformation of the surface, the value of $\Sigma\alpha$ will in general be altered; whereas the expressions for q, t should it is clear remain unaltered; and it arises as follows, viz. for certain transformations of the surface the curve of the order $(k - n + 4)n' - 3$ which is the transformation of the residue (as defined by the foregoing conditions) ceases to be a curve of the assigned order, and to satisfy the order conditions; this is the assumed manner; and then the coefficient $(a, \dots)$ will be the like functions of the elements an indecomposable curve but contains a certain number ω of factors (each belonging to a unicursal curve definable by means of the number of its passages through the several points a_i), which factors are to be rejected in order to obtain the equation of the proper residue. Thus reverting to the transformation

$$x : y : z : w = x' : y' : z' : (x' + y' + z')^2$$

of Steiner's surface, the projection of the quartic residue was (as already remarked) a line; applying to the plane figure the ordinary quadric (or inverse) transformation we introduce three fixed points, ($n_1 = 3$), say these are A, B, C ; viz. in the new transformation of the surface the projection of the residue ought clearly to be a conic through the three points; but according to the general formula it is a quintic having at each of the points a triple point: the quintic in question is consequently made up of the conic and the three lines $1A, 1B, 1C$; that is, it is on this account that a like explanation applies to the first investigation of $2q + 9t$.

I further remark, reverting to the equations

$$x : y : z : w = X' : Y' : Z' : W'$$

of the transformation, that the product of the ω factors is given as the common factor (if any) of the Jacobians

$$J(Y', Z', W'), \quad J(Z', W', X'), \quad J(W', X', Y') \quad \text{and} \quad J(X', Y', Z').$$

Such common factor exists whenever we can by a Cremona-transformation of the plane figure reduce the number of the points a , upon which the transformation of the surface depends; viz. for any given transformation of the surface, ω is equal to the excess of $\Sigma\alpha$ above the minimum value of $\Sigma\alpha$, and in this independent of the particular transformation. And of course of $\Sigma\alpha$ has this minimum value, viz. if the transformation is such that the number of the points a , cannot be reduced by any Cremona-trans-

p. 393

formation of the planar figure, then we have $\omega = 0$. I presume that for the most simple transformation, that is, when σ' has its least value, $\Sigma\alpha$ has also its least value, and consequently that $\omega = 0$.

Recapitulating, the results obtained are

$$q = -2(n - 5) - 1 + 4n^2 + 4n^2 - 18 + 4(\Sigma\alpha - \omega),$$

$$t = (n - 5) - \frac{1}{2}n^2 - 4n^2 + \frac{1}{2}n - 4(\Sigma\alpha - \omega),$$

where δ is to be excluded from

$$\delta = \frac{1}{2}(n - 2)(n - 3) - b + p,$$

the formulae are verified in the several cases:

*[Table: 6 column table with $n, n', \Sigma\alpha, \delta, t, q, p$ and entries for Quadric surface, Cubic scroll, Quartic with nodal conic, Quartic with nodal quadriquadric, Quartic with two nodal lines etc.
— table omitted]*

which are the transformations chiefly as yet examined: but the first-mentioned case (quadric surface, generalised stereographic projection), although as stated the formulae are verified with the value $\omega = 1$, does not really come under the foregoing theory. It is interesting to see that they are verified in the last-mentioned cases, belonging to a negative value of D , that is, to a special system of formulæ.

Cambridge, Dec. 5, 1870.

[524]

On the Deficiency of Certain Surfaces.

[From the *Mathematische Annalen*, vol. III. (1871), pp. 526–529.]

p. 394

If a given point or curve is to be an ordinary or singular point or curve on a surface of the order n , this imposes on the surface a certain number of conditions, which number may be termed the “Postulation”; thus “Postulation of a given curve ρ -tuple curve on a surface n ” will denote the number of conditions to be satisfied by the surface in order that the given curve may be a ρ -tuple curve on the surface.

The “deficiency” (Flächenpoldecel) of a given surface of the order n is

$$= \frac{1}{6}(n-1)(n-2)(n-3) - \text{the deficiency-value of the several singularities; viz.}$$

as shown by Dr Noether, if the surface has a given ρ -tuple curve, the deficiency-value hereof is

$$= \text{Postulation of the curve } (\rho-1)\text{-tuple curve on a surface } n-4;$$

and if the surface has an ordinary singular point, the deficiency-value hereof is

$$= \text{Postulation of the point } (\rho-1)\text{ conical point on a surface } n-4,$$

viz. this is $= \frac{1}{6}(t-1)t(t+1)$ where, as I write throughout, $t = \rho - 3$.

I remark that if the tangent-cone at the conical point has δ double lines and ν cuspidal lines, then the deficiency-value is

$$= \frac{1}{6}(t-1)t(t+1) - \frac{1}{2}\delta t - \nu(t-1) - (\text{etc.})$$

In the case of a double or cuspidal curve δ a ρ -tuple curve on the surface n has the deficiency-value = Postulation of given curve (ρ) simple curve on a surface $n-4$; and so for an ordinary cuspidal point: $\nu = 2$, and the deficiency-value = 0; results which were first obtained by Dr Clebsch.

p. 395

I found in this manner the expression for the deficiency of a surface having a double and cuspidal curve and the other singularities considered in my "Memoir on the Theory of Reciprocal Surfaces," *Phil. Trans.*, vol. CLI. (1869), [411, Coll. Math. Papers, vol. VI. p. 358]; viz. this was

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-4)b + \frac{1}{6}(b+r) + 2t + \frac{1}{2}\beta + \frac{1}{2}\gamma + \frac{1}{2}n - \frac{1}{2}b,$$

where we have

b , order of double curve,

h , class of Do ,

δ , order of cuspidal curve,

β , number of intersections of the two curves, stationary points on b ,

γ , number of intersections, stationary points on r ,

j , number of intersections, not stationary points on either curve,

θ , number of certain singular points on b , the nature of which I do not complete to understand it is here taken to be $= 0$.

Before going further I remark that

Postulation of right line god β -tuple on surface n

$$= \frac{1}{6}(t+1)(n-t) - \frac{1}{6}t(t+1)(2t-5),$$

or we have

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - \frac{1}{4}t(t+1)(2n-5),$$

or we have

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - \frac{1}{4}t(t+1)(2n-5),$$

so that $D = 0$ either if $n = t - 1$ or if $n = t$, viz. these are surfaces of a scroll (skew surface) with a $(n-1)$ -tuple right line, the latter that of a surface with a $(n-2)$ -tuple line: those are known to be of deficiency $D = 0$.

For a surface of the order n with a conical point where the tangent-cone has δ double lines and ν cuspidal lines, the deficiency-value is

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - \frac{1}{6}(t-1)t(t+1) - \frac{1}{2}\delta t + \nu(t-1)$$

viz. for $n = t$ this is $D = 0$. In fact, a surface with a $(n-1)$ -tuple conical point is once seen to be rationally transformable into a plane; and for $r = t$, that is, for a cone of the order n , we have

$$\begin{aligned} D &= \frac{1}{6}(n-1)(n-2)(n-3) - \frac{1}{6}(n-2)(n-3)(n-4) - \frac{1}{4}(n-3)(n-4) \\ &= \frac{1}{4}(n-1)(n-2) - \frac{1}{4}n(n-3), \end{aligned}$$

where the last term $-\frac{1}{4}\delta(n-4) - \nu$ is added because in the present case the surface has the δ double lines and the ν cuspidal lines.

p. 396

The formula therefore gives

$$D = -\frac{1}{6}(n-1)(n-2) + \delta;$$

viz. this is equal to the deficiency of the plane sections taken negatively.

I find that the same property exists for the deficiency of the planes sections taken negatively, having only a double curve; and *actually* in the case of a torse (developable surface) having a cuspidal curve with the ordinary singularities; and this being so there can I think be no doubt that for it is true for any small or torse whatever—viz. that in the face of the tetrahedron, or any in the plane of an equilateral triangle, the sum of any ruled surface whatever the deficiency is equal to that of the plane section taken negatively.

First, for the scroll, we have

$$D = \frac{1}{6}(n-1)(n-2)(n-3) - (n-4)b + \frac{1}{6}\beta + \frac{1}{2}\beta,$$

which should be

$$= -\frac{1}{6}(n-1)(n-2) + b,$$

this is equal to the deficiency of the planes sections taken negatively.

Secondly, for the torse; changing the notation into that used by the inequalities of the curve and torse, we have

$$\begin{aligned} D &= \frac{1}{6}(n-1)(n-2)(n-3) - (n-4)x + n + \frac{1}{6}(\theta + r) + 2\beta + \frac{1}{6}\beta + \frac{1}{2}\gamma + \frac{1}{2}\beta, \\ &= -\frac{1}{6}(n-1)(n-2) + a. \end{aligned}$$

We have $g = r(n-8) - 8x$, and substituting this value and expressing everything in terms of r , u , z by means of the formulae

$$\begin{aligned} n &= \frac{1}{4}(r - n - 2a), \\ a &= a - 3\delta - 2x - 3\kappa + 4\beta + 2\kappa, \\ g &= a + (n-2)x - 2\tau\kappa, \\ g &= a - n + (n-2)z - 2\kappa\kappa, \\ n &= a + (n-2)z - 6g, \\ b &= \frac{1}{4}(n-2)x - 2g, \end{aligned}$$

we have after all reductions

$$D = -\frac{1}{6}(n-1)(n-2) + r - 1 = -(n-1)(n-2) + \delta.$$

p. 397

We have then a class of surfaces of negative deficiency; viz. any rational transformation of a cone in the plane section has a given (positive) deficiency produces such a surface; and I think it must be conversely that a surface of negative deficiency is always the rational transformation of a cone for which the deficiency is equal to that of the surface taken with the reverse sign; viz. in the sentence, the formula for the deficiency is $(n - 1)$, $b = 2$, $\beta = 2$, $\theta = 0$, $i = 0$.

$$D = \frac{1}{6}(n - 1)(n - 2)(n - 3) - (n - 4)b + \frac{1}{6}(b + r) + 2(n - 5) + 2(n - 1).$$

viz. a scroll surface is obtained as the quartic inverse of a cubic cone; viz. taking for the vertex the point $x : y : z = 0$, w may be a cubic function of x, y, z , and the remaining quartic also: of: $x : y : z$ and the equation of the cone is

$$(ax - mn, by - py + \theta, kx - jz)^2 = 0,$$

where (A, B) etc. = 0.

Taking Q a quartic function (x, y, z) , the transformation in question consists in the change of x, y, z, w into xQ, yQ, zQ, wQ . (where Q is inferior or equal to the plane $x, y, z = 0$ and having Q as the equation of the cone in question and the conic $Q = 0$ as nodal conic $w = 0, Q = 0$).

Cambridge, 5 Jan., 1871.

[525]

An Example of the Higher Transformation of a Binary Form.

[From the Mathematische Annalen, vol. IV. (1871), pp. 359–361.]

p. 398

THE quartic

$$(1) \quad (a, b, c, d, e)(x, y)^4$$

is by means of the two quantics

$$(T) \quad (a, B, y)(x, y)^2 \quad \text{and} \quad (a', B', y')(x, y)^2$$

transformed into

$$(2) \quad (a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4,$$

that is, eliminating x, y from

$$(a, b, c, d, e)(x, y)^4 - x(a, b, y)(x, y)^2 + y(a', b', y')(x, y)^2 = 0,$$

we obtain

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = 0.$$

It is required to express the invariants of (2) in terms of the simultaneous invariants of (1) and (2).

Write

$$P, Q, R = ax, +by, +\gamma y_1, \quad a'x_1 + B'y_1 + \gamma'y_1.$$

the equations from which (x, y) have to be eliminated are

$$(a, b, c, d, e)(x, y)^4 = (P, Q, R)(x, y)^2 = 0,$$

we obtain

$$(a, b, c, d, e)(x, y)^4 = (P, Q, R)(x, y)^2 = 0.$$

p. 399

and the result of the elimination therefore is

$$\begin{vmatrix} a, & 4b, & 6c, & 4d, \\ e & & & \\ & a, & 4b, & 6c, \\ 4d, & e & & \\ & & P, & 2Q, \\ R & & & \\ & & & P, \\ 2Q, & R & & \\ & & & \\ P, & 2Q, & R & \end{vmatrix} = 0,$$

viz. this determinant is the transformed quartic $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$.

The developed expression of the determinant is

$$\begin{aligned} & a^2D - 8bQR \\ &= \begin{pmatrix} -12ac \\ +16b^2 \end{pmatrix} P^2R^2 - 24cRQ^2PR + \begin{pmatrix} 24bc \\ -64bd \end{pmatrix} PQ^2R - 32cdQ^4R \\ & \quad + \begin{pmatrix} -32cd \\ +64bd \end{pmatrix} P^2R^2 - 32cdQ^2PR + 16c^4 \\ & \quad + \begin{pmatrix} 32d^2 \\ -64cde \end{pmatrix} PR^3 - 32deQR^3 + 16e^4 \\ & \quad + \begin{pmatrix} 24de \\ +64cd \end{pmatrix} PR^2 - 24eQR^2 + 16d^4 \\ & \quad - 4bP^3Q + eP^3. \end{aligned}$$

or that writing for P, Q, R their values, we have the transformed function $(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4$; the coefficients being of the form

$$a_1 = (a, b, c, d, e)(a, \beta, \gamma)^4, \quad (\alpha', \beta', \gamma')^4.$$

$$b_1 = \&c., \quad \&c., \quad \&c.$$

$$c_1 = \&c., \quad \&c.$$

$$d_1 = \&c., \quad \&c.$$

$$e_1 = \&c., \quad \&c.$$

Writing I, J for the invariants of the quartic (1), and

$$A = b(c\delta^2 - 4B\gamma) + c(\gamma a - \beta^2) + d(a\beta - a\gamma),$$

$$B = ac, \quad a, \quad b, \quad c, \quad d, \quad b, \quad c, \quad \beta - \beta^2,$$

A, B simultaneous invariants of the forms (1) and (2). Putting more- over $\nabla = P^2 - 2QR$, and writing I, J the like invariants of the form (5), and

$$J_1 = bI(B - IA) + B(IAB - 2aB)AB + (3DB - 3aB)A^2,$$

$$J_2 = b(aB - IA)AB + B(IAB - 2aB)AB + (3DB - 3aB)A^2,$$

we have

$$\nabla_1 = 256\nabla^2(I^2\nabla - IAB)\nabla.$$

p. 400

As a verification, suppose $(\alpha, \beta, \gamma) = (1, 0, c)(1, 0, c)^2 = c + cx^2$ and $\nabla = P^2 - 2QR = -4bc$. Take also $(a', \beta', \gamma') = (0, 1, b)(1, 0)^2 = y$; the transforming equation is then $I = 4J$, and $(X, Y, x) = (-1, 0, 1)$. We have $\nabla = P^2 - 2QR$, $-4bc$, $+b \cdot y^4 \cdot a$, and $G = ac$, $D = 16$, and $(X, Y, x) = (1, 0, 1)$. We have

$$\text{Det.} = (P, P^2(W + 2YPW^2)) = 2Q + (a + 2y)w - 2y^4 = (a + 2y, b, ay + 2y)^4 = 0.$$

that is,

$$(a_1, b_1, c_1, d_1, e_1)(x_1, y_1)^4 = -2x_1 - 2(a + 2y)x_1^3y_1 + 6ayx_1^2y_1^2 + 2(4ay^2 - 2y^4)y_1^3.$$

whence

$$I_1 = \frac{4096}{3} \cdot a^2b^2c^2(4 + (2b)^2 \cdot 256),$$

$$J_1 = \frac{262144}{27} \cdot b^4,$$

also

$$A = -16, \quad B = 8,$$

and the equations for I_1, J_1 become

$$\frac{4096}{3} = 4\delta(b + 2 \cdot 256),$$

$$\frac{262144}{27} = b\{-16 \cdot (64 + \delta \cdot 4096), \}$$

which are true. More generally, assuming

$$(a, b, c, d, e) = (4\delta_0, c, x_0 + 6\delta b)(x_1, y_1)^4 = 0$$

(where $I = 1 + 24\delta^2, J = \delta - \delta^4$), we have the same transforming equation; we have

$$\nabla = P^2 - 2QR, \quad ec, \quad c, \quad a^4 - 2bc, \quad b, \quad a, \quad c, \quad b, \quad b, \quad b, \quad c, \quad b\delta;$$

whence

$$I_1 = \frac{27}{3}(1 - 24\delta^2), \quad J_1 = \frac{a^4}{27}(1 - 244\delta^2),$$

also

$$A = -16, \quad B = 8 + 1.$$

Substituting these different values in the equations for I_1, J_1 , we obtain

$$16(1 - 24\delta^2) - (1 + 240\delta) - 768 \cdot (75 - 56\delta) + 49 \cdot 4(1 + 24\delta),$$

and

$$-3(3 - 540\delta) + 27 \cdot 49\delta - 49 \cdot 4(1 + 240\delta) + 27 \cdot (1 + 240\delta) - 12 \cdot 24 \cdot 4\delta - 16 \cdot (1 + 248\delta) + 4 \cdot 4(1 + 24\delta),$$

which are in fact satisfied identically.

Cambridge, 26 July, 1871.

[526]

On a Surface of the Eighth Order.

[From the *Mathematische Annalen*, vol. IV. (1871), pp. 558–560.]

p. 401

I REPRODUCE in an altered form, so as to exhibit the application thereto of the theory of the six coordinates of a line, the analysis by which Dr Hierholzer obtained the equation of the surface of the eighth order generated by the lines of the vertex of a quadricone which touches six given lines.

I call to mind that if (A, B, C, D, E, F, G) are the coordinates of any two points on a line, then the quantities (a, b, c, f, g, h) , which denote respectively

$$(by' - y'a, az - a'a, cy' - y'a, cz' - z'a, az - a'a, az' - z'a),$$

and which are such that $af + by + cz = 0$, are the six coordinates of the line^{*};

Consider the given point (x, y, z, w) and the lines (a, b, c, f, g, h) , and write for shortness

$$\begin{aligned} P &= hy - gx + aw, \\ Q &= -ax + fx + bw, \\ R &= gx - fy + cw, \\ S &= -ax - by - cz, \end{aligned}$$

then taking (X, Y, Z, W) as current coordinates, the equation of the plane through the given point and line is

$$PX + QY + RZ + SW = 0,$$

Considering in like manner the given point (x, y, z, w) and the three lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \dots)$, $(a_3, \dots)$, then we have the three planes

$$\begin{aligned} P_1X + Q_1Y + R_1Z + S_1W &= 0, \\ P_2X + Q_2Y + R_2Z + S_2W &= 0, \\ P_3X + Q_3Y + R_3Z + S_3W &= 0. \end{aligned}$$

^{*} Cayley, "On the six coordinates of a line," *Camb. Phil. Trans.* vol. XI. (1868), [435], pp. 290–323.

p. 402

and if these planes have a common line, the point (x, y, z, w) is in a line meeting each of the three given lines; that is, in the locus of the points in the hyperboloid through the three given lines. It follows that the equations

$$\begin{vmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{vmatrix} = 0$$

reduces themselves to a single equation, that of the hyperboloid in question.

I write for shortness

$$\begin{aligned} (000) &= (a\beta\delta)x^2 + (bb\gamma)y^2 + (cfy)y^2 + (ch\beta)x^4 \\ &+ [(a\beta\zeta) - (abh)]yx \\ &+ [(c\beta\eta) - (cbf)]zy \\ &+ [(c\gamma g) - (abc f)]xy \\ &+ [(bfg) + (afh)]yw \\ &+ [(bfg) + (ach)]xz \end{aligned}$$

viz. (123) will mean $(a_1a_2a_3)x^2 + \text{etc.}$ where $(a_1a_2a_3)$ denotes as usual the determinant

$$\begin{vmatrix} a_1 & \beta_1 & \delta_1 \\ a_2 & \beta_2 & \delta_2 \\ a_3 & \beta_3 & \delta_3 \end{vmatrix} \text{ etc.};$$

then the equation in question are found to be $(123) x^2 = 0$, $y(123) x = 0$, $z(123) z = 0$, $w(123) w = 0$, reducing themselves to the single equation $(123) = 0$, which is accordingly that of the hyperboloid through the three lines*.

Proceeding now to the above-mentioned problem, we have the point (x, y, z, w) , and the six lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, \dots)$, $(a_6, \dots)$; the lines 1, 2, 3, 4, 5, 6: the six planes

$$P_1X + Q_1Y + R_1Z + S_1W = 0, \text{ etc.}$$

must be tangents to the same quadricone; that is, considering the sections by the plane $W = 0$, the six lines

$$P_1X + Q_1Y + R_1Z = 0, \text{ etc.}$$

must be tangents to the same conic; and the condition for this is

$$[123456] = 0$$

* This equation is given in the paper above referred to, p. 314.

p. 403

where the symbol stands for the determinant

$$\begin{vmatrix} P_1, & Q_1, & R_1, & Q_1R_1, & R_1P_1, & P_1Q_1 \\ P_2, & \dots & & & & \\ \dots & & & & & \end{vmatrix} = 0.$$

But as is well known this equation may be written

$$(*) \quad (126)(845)(145)(256) - (146)(125)(345)(245) = 0,$$

when (126) etc. denote the determinants

$$\begin{vmatrix} P_1, & Q_1, & R_1 \\ P_2, & Q_2, & R_2 \\ P_6, & Q_6, & R_6 \end{vmatrix} \text{ etc.};$$

or, what is the same thing they denote the functions above represented by the like symbols (126) viz. $(a_1a_2a_6x^2 + \text{etc.})$ The equation (*) just obtained is therefore the for the surface of the eighth order, the locus of the vertex of a quadricone which touches six given lines.

I remark that in my "Memoir on Quartic Surfaces," *Proc. Lond. Math. Soc.*, vol. III. (1870), [445], pp. 19–69, I obtained the equation of the surface which the foregoing form $[123456] = 0$ or any $[(P, Q, R)] = 0$, taking that there a factor a^* , so that the order of the surface is $= 8$; and further that the equation $\beta y + bz = (-ax + \beta y - bz)\nu'[ax + \beta y - bz](-ax + \beta y - bz) = 0$, where exp. ω (read exponential) denotes a^*x^x and $[(a, b, ce)]$ denotes

$$\begin{vmatrix} a, & b, & c \\ b, & b_1, & c_1 \\ c, & c_1, & b_1 \end{vmatrix}$$

Also that the equation contains the four terms

$$a^2[ab, -b, gh]x'^2 - [abf]x'^2(b - f), \quad a^* + ay^2[-f, g, cf]^* = -a[a, b, cf]^*.$$

Cambridge, 12 September, 1871.

[527]

On a Theorem in Covariants.*[From the Mathematische Annalen, vol. v. (1872), pp. 625–629.]**p. 404*

THE PROOF given in Clebsch “Theorie der binären algebraischen Formen” (Leipzig 1872) of the finite number of the covariants of a binary form depends upon a subsidiary proposition which is deserving of attention for its own sake.

I use my own hyperdeterminant notation, which is as follows: Considering a function $U = (a, \dots)(x, y)^n$ (viz. $U_1 = U_2 = \dots = (a, \dots)(x_1, y_1)^n$ and writing $12 = d_{x_1}d_{y_2} - d_{y_1}d_{x_2}$, then the general form of a covariant of the degree θ is

$$12^{r_{12}} 13^{r_{13}} 23^{r_{23}} \dots U_1 U_2 \dots U_\theta;$$

where A is a sound numerical factor, the indices $a, \beta, \dots$ are positive integers, and after differentiations (each set of variables $(x_1, y_1) \dots (x_2, y_2) \dots$ etc., is changed into (x, y) . In the general form of a covariant is as above, viz. a covariant is equal to a sum of terms of the form $A, 12^a \dots U_1 U_2 \dots U_\theta$, but writing the term, before A is equal to a sum of terms of the form $A, 12^{a_{12}} \dots U_1 U_2 \dots U_\theta$, but writing the term, before $U_2 \dots$ etc., the symbols which need to be replaced by (x, y) for distinctness; the several sets of variables are each replaced by (x, y) in all the orders—one of these functions with ∇ , and so for the other order—this can be obtained by interchanging the symbols. $\nabla, \nabla', \nabla'', \nabla'''$, then by means of the relation

$$\nabla = 12 U_1 U_2, \quad \nabla = 12 \cdot 13 \cdot U_1 U_2 U_3 = (12 \cdot 13) \text{ etc.}$$

being now in the first instance a function of the single set (x, y) of variables.

p. 405

We may call the symbol $12^\alpha 13^\beta 23^\gamma, \dots$, and consider any of the symbols

$$k \cdot 1 \overline{12}^\alpha \overline{13}^\beta \overline{23}^\gamma \dots, \quad \text{or} \quad k \cdot 2^\alpha \dots, \quad \nabla_{\alpha \dots \theta} = k (\overline{12} \overline{13} \overline{23} \overline{14} \dots) \nabla,$$

which, under either of the two forms, $[k]$, denotes one of the symbols $[k], \dots$ etc., that this is considered as a symbol involving the θ symbolic numbers $1, 2, 3, \dots, \theta$, even although in pursuance of the rule above stated, the symbol involving the actual expression of the symbol does not contain these symbolic numbers, viz. may have $x = w, y = u, \dots, y = e$. This being the case, ∇ is a covariant of U , but several sets of variables are each replaced by (x, y) in all the orders—one of these functions $\nabla, \nabla', \nabla''$, then by means of the relation

$$\nabla = 12 U_1 U_2.$$

For the four numbers $1, 2, 3, 4$ we have a group of the 15 formulæ

$$\nabla, 12 - 34; \nabla, 14 - 23; \nabla, 33 - 14 = 0,$$

$$\nabla, 24 + 13 \cdot 12 - 14, 23 = 0,$$

$$\nabla, 23 + 14 - 12, 34 = 0,$$

leading to

$$23. 14 + 31. 24 + 12. 34 = 0,$$

which is a form not involving the ∇ 's and consequently is applicable to the transformation of invariant-symbols where the numbers

$$a_1, a_2, \dots, a_n, \dots$$

are all = 0.

I establish the following definitions:

A symbol $[12 \dots m]$ is prominent when each index-sum is $\leq c$; otherwise it is ultimate; viz. this is the case when any one or more of the index-sums is or are $> c$. A symbol $[12 \dots m]$ may be said to be *primary* when it is $= w$; and that this is the ultimate as to $1, 2$ if a, c , and so on; and also as to a row when in other cases.

A prominent symbol then may any index-sum thereof is $\leq c$, is unable to *inferior*: thus if $w \geq c$, the symbol is inferior in regard to 1; and if $w \geq c, \geq c, \text{ult} \geq c$, ult is inferior in regard to 1 and 2: and like for the other cases.

p. 406

degree by a number which is greater than $\frac{1}{2}n - 1$; in fact, suppose θ inferior in regard to 1, then the order is

$$\theta n = w_1 + (n - w_2) + \dots + (n - w_n),$$

where each term after the first is at least $= \frac{1}{2}n$, and the order is at least $= w_1 + \frac{1}{2}n(\theta - 1)$, hence order — degree is at least $= w_1 - \frac{1}{2}n$, w_1 being less than $\frac{1}{2}n$, this is greater than $\frac{1}{2}n - 1$.

Conversely, if for any symbol order-degree is $\leq \frac{1}{2}n - 1$, the symbol is not inferior.

A symbol $[12 \dots m]$ is called inferior in regard to a row when it is $= w, \dots$, in this it is when each index is w , or sharp when each index w .

The subsidiary theorem is now as follows: A symbol is inferior or sharp; or each index-sum being νc a primary symbol, the order of any symbol which is sharp or inferior, expressed in terms of the order and the index-sums of the same number of terms each of which is inferior or sharp: viz. as to the order, the symbol is not sharp, but no stationary points, so that q being the class, we have $q = D - k - 2\delta - 3\nu$; and $z = w$.

The proof $[12 \dots m]$ is inferior, suppose it is to be in regard to 1, then there is

$$\overline{12}, \overline{14}, \overline{2} \cdot \overline{14}.$$

this expression by a denominate νn is not $= w$, and we may then by means of terms each of which is inferior or sharp: we may then by means of the form $\nabla', \nabla'', 12^{14} 23^{18}, \overline{23}$, then by means of the relation

$$\nabla', \nabla' = w_1 - 1, \quad w_2 = w_2 - 1, \quad w_3 = w_2 - 1,$$

$$Q = \nabla', \nabla', \dots, \nabla', \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots, \nabla' \dots$$

then supposing for a moment that 1, is not $= w$, and to substitute the expression contains the factor ∇', ∇' , which is equal to and may be replaced by $\nabla', \nabla', \nabla' \nabla'$, in the second side of variables respectively.

$$Q = \nabla' \nabla'.$$

and

$$\Omega \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla' \nabla'.$$

Now for Ω the sum of the indices $1_n - 1, 2_2 - 1, 2_2 - 1$, so that by hypothesis Ω is inferior or sharp: 1_n is in inferior or sharp, since one of these is sharp by hypothesis prove that Q is inferior or sharp, we have just been to prove that Q in a way also that the inferior or sharp.

p. 407

expressed as a sum of terms each of which is inferior or sharp; and we have to prove this for the next following degree; $u + 1$, $u + 1$, or the conveniences p in place of $m + 1$, (say for the degree p); that is, for a symbol

$$[12 \dots mp] = p_1^{r_1} p_2^{r_2} \dots p_m^{r_m} [12 \dots mp].$$

I write as before, $\sigma_1, \sigma_2, \dots, \sigma_m$ for the index-sums of $[12 \dots mp]$; therefore as $\sigma_1 + \sigma_2 + \dots + \sigma_m$, $\sigma_1, \dots$, are the index-sums of $[12 \dots m]$ involving $p_1, p_2, \dots, p_m$.

If $[12 \dots m]$ is sharp, then $[12 \dots mp]$ is inferior in regard to p .

If $[2, m]$ is inferior, suppose it is so in regard to p ; then we have $\sigma_1 = \text{sharp}$ (being therefore inferior) and when $\sigma_1 = \text{no}$. In the case of any one of the indices being negative, viz. $p < r$, this is sharp; but in any other case, $[12 \dots mp]$ being inferior in regard to p .

Consider the expression

$$[12 \dots mp][12 \dots pp] \dots p[12 \dots m],$$

where, $\sigma_1, \sigma_2, \sigma_3, \dots$ are in inferior, and when $\sigma_1, \dots, \sigma_m$, has any one of these is $\sigma_1, \dots, \sigma_m$ are sharp negative.

Assume that when $[12 \dots mp]$ is inferior, and when $\sigma_1, \sigma_2, \dots, \sigma_m$, have any values such that their sum is greater than a given value σ , the expression is a sum of terms each of which is inferior or sharp: we wish to show that this also is, so the last term not exceeding value, $= \sigma$, the case will then be the same.

For this purpose, introducing the ∇ 's I write

$$Q = \nabla_{12} \nabla_{13} \nabla_{23} \dots \nabla_{1p}^{r_1} \dots \nabla_{2p}^{r_2} \dots \nabla_{mp}^{r_m} [12 \dots m],$$

then supposing for a moment that σ_1 is not $= 0$, and to substitute the expression contains the factor $\nabla_p \nabla_p$, which is equal to and may be replaced by $\nabla_p \nabla_p$, $\nabla_p \nabla_p$, in the indices of which the index 1 is one less, and the index $1 - p$ one greater.

We have then

$$Q = \nabla_{12} \nabla_{13} \dots \nabla_{1p}^{r_1} \dots \nabla_{2p}^{r_2} \dots \nabla_{mp}^{r_m} [12 \dots m]$$

and

$$\Omega = \nabla_{12} \nabla_{13} \dots \nabla_{1p}^{r_1-1} \dots \nabla_{2p}^{r_2+1} \dots \nabla_{mp}^{r_m} [12 \dots m].$$

Now for Ω the sum of the indices $\sigma_1 - 1, \sigma_2, \sigma_3, \dots$ as the same value, ν . And the symbol thus obtained is, by hypothesis Ω is inferior or sharp; in $12^*, 13^*$, etc., the index-sums equal to the prove that Q is inferior or sharp, Ω has ν inferior or sharp; we have, by hypothesis, Ω is inferior or sharp.

p. 408

which is diminished by unity. Such change is possible so long as the index 1, has not attained its maximum value, $= a$, or $= a$; or as the case may be, and there is any other index $1_a, 1_b, \dots$ which is not $= a$; that is, we may pass from Q to Ω , from Ω to Ω' , and so on; and it will be sufficient to show that the last terms of the series is inferior or sharp. We thus pass from Q to R , where

$$R = \rho_1^{r_1} \rho_2^{r_2} \dots \rho_m^{r_m} [12 \dots m].$$

and $a_1, a_2, \dots, a_m$ $a_1 + \dots + a_n - 1$, so are also a.

According as $a > c$, or $w < c$, R is inferior or sharp.

Now let Ω the sum of the indices $1_a, 1_b, \dots$ be greater than a , that is, by hypothesis Ω is inferior; we let σ_1 greater than σ_2 . Then if σ_2 be less than σ_1 , we let σ_1 greater than σ_2 . Then σ_2 becomes the last formed term which is inferior in regard to p_1 ; ult is, for in the indices of Ω the last mentioned form, $a_p = a_1 + a_2 + \dots$ becomes the value σ_2 , which is equal to and may be replaced by greater than a_p .

So that comparing with the Ω ult, it has, but no two of the inferior in regard to p_1 , that is, Ω is inferior, $\sigma_2 = a$, $-\sigma_1$, and correct a given value $\sigma_1 = a$, $-\sigma_2$, the theorem is true for the next succeeding value σ_1 , or being true for the case $\sigma_1 = a$, $-\sigma_1$, it is true generally.

Cambridge, 24 April, 1872.

[528]

On the Non-Euclidian Geometry.*[From the Mathematische Annalen, vol. v. (1872), pp. 630–634.]**p. 409*

THE theory of the Non-Euclidian Geometry as developed in Dr Klein's paper "Ueber die Nicht-Euklidische Geometrie" may be illustrated by showing how in such a system we actually measure a distance and an angle: and by establishing the trigonometry of such a system. I confine myself to the "hyperbolic" case of plane geometry: viz. the absolute is here a real conic, which for simplicity I take to be a circle: and I attend to the points within the circle.

I use the simple letters $A, B, \dots$ to denote (Anschauung points or angular) distances measured in the ordinary manner: and the same letters, with a circumflex $\hat{A}, \hat{B}, \dots$ to

[Figure: circle with inscribed triangle and various line segments — diagram omitted]

denote the same distances measured according to the theory. The radius of the absolute is for convenience taken to be $= 1$: the distance of any point from the centre can therefore be represented as the sine of an angle.

p. 410

The distance BC , or any θ , of any two points B, C is by definition as follows: Radius of circle = 1.

$$\begin{aligned} 1\circ. \quad & ABC, \text{ sides are } a, b, c : \\ & \text{angles } A, B, C : \\ & OA, OB, OC : = \sin g, \sin g, \sin g : \\ & OA', OB', OC' = \sin a, \sin b, \sin c : \\ & \angle BOC, COA, AOB : = b, c, B : \\ & \hat{a} = \frac{1}{2} \log \frac{BJ.CI}{BI.CJ}. \end{aligned}$$

(where I, J are the intersections of the line BC with the circle); that is,

$$e^x = e^x = \frac{BJ.CI}{BI.CJ} = \frac{BJ.CI}{OB^2.OJ} \cdot \frac{BJ.CI}{OB^2.OJ} \cdot \frac{BJ.CI}{OBA} \cdot \frac{CO.CJ}{OB.CJ} \cdot \frac{CO.CJ}{OBA}.$$

where the numerator is

$$\begin{aligned} BI(BJ - BC) + CI(BC + CJ) &= BC.BJ + CI.CJ + BC(CI - BJ); \\ &= BC.BJ + CI(CJ + BC) = BC.BJ + CI(CJ + BC); \\ &= BC.BJ + CI(CJ + BC); \end{aligned}$$

Hence taking α for the angle whose sine is BC , and for the distances OB, OC respectively, we have $BJ.BI = \cos^2 \theta$, $CI.CJ = \cos^2 \theta$; and for the distance BC of these distances B, C as the perpendicular distance from O on the line BC . $BJ - BC + CJ$ as the perpendicular distance from O on the line BC . $BI = BC - C - BI = 2c \cos \alpha + a \cos \alpha$, or, what is the same thing, taking α for the angle BOC , and therefore

$$a' = a \sin \theta = a \cos \alpha - \cos \alpha \theta - \cos \alpha \theta - \cos \alpha \theta,$$

we have

$$\cosh \hat{a} = \frac{1 - \sin a \sin b \cos C}{\cos a \cos b}.$$

In a similar manner, if also α is the perpendicular distance from O on the line BC (that is, $a \sin a = \sin \theta \sin a \cos a$) it can be shown that

$$\sinh \hat{a} = \frac{a \cos a}{\cos a \cos b},$$

the equivalence of the two formulas appearing from the identity

$$\cos^2 \alpha \cos^2 \theta = (1 - \sin \alpha \sin a \cos a)^2 = a \sin^2 \alpha \sin^2 \theta;$$

which is at once verified.

Next for an angle; we have by definition

$$\hat{A} = \frac{1}{2i} \log \frac{AK.AL}{AL.AL}$$

p. 411

where AI, AJ are the (imaginary) tangents from A to the circle; or writing for shortness BI for, instead of AKI, AKJ , (the angular point being always at A), consequently

$$\hat{A} = \frac{1}{2i} \log \frac{\sin BI \cdot \sin CJ}{\sin BJ \cdot \sin CI},$$

consequently

$$e^{i\hat{A}} = e^{i\hat{a}} \sqrt{\frac{\sin BI \sin CJ}{\sin BJ \sin CI}},$$

where the numerator is

$$BI(BJ + BC) - CI(BC - CJ) = (BJ - CJ)BC + \sin BJ \cdot BC + \sin BJ \cdot CI + BI \cdot CI,$$

or say

$$e^{i\hat{a}} = \sqrt{\frac{\sin BJ \sin CI}{\sin BJ \sin CI}},$$

where $BJ \cdot BI = \cos^2 AB$, $CI \cdot CJ = \cos^2 AC$, (the angular point being always at A), moreover

$$\sin BJ \cdot CI = e^x = \cos AB \cdot \cos AC \cdot \sin BC,$$

the perpendicular distance from O on the line BC is $= \sin BC$; moreover

$$\sin BJ \cdot CI - \sin BJ \cdot CI = 2 \sin a \cos b \cos ac - \sin BC = -2 \cos AB \cdot \cos AC \cdot \sin BC,$$

and $\sin BJ \sin CI = a^2 \sin BC + a^2 \cos BC$.

$$\sin BJ - \sin BJ \cdot CI = 2 \sin a \cos b \cos ac - \sin BC = -2 \cos AB \cdot \cos AC \cdot \sin BC,$$

in CI sub $AJ = 2 \sin$, in C $\sin ac \sin BC$; and similarly $\sin BJ \cdot CI = \cos ac \sin ac$ in BC ,

$$\sin BJ + \sin CI = 2 \sin a \cdot a = a \cdot a \sin \alpha.$$

whence the required formula

$$\sin \hat{A} = \frac{a \cdot p \sin A}{\cos b \cos c}.$$

In the same way, or analytically from this value, we have

$$\cos \hat{A} = \frac{1 - \sin a \cos b \cos c}{\cos b \cos c},$$

and thence also

$$\tan \hat{A} = \frac{a \cdot p \sin A}{1 - \sin a \cos b \cos c}.$$

In particular, taking the line AC to pass through O , we will have $b = O$, that is, we have $\sin BO = \cos a \sin BC = b$, that is, $BC = \sin^{-1} a$, $\cos a \tan b$, and writing $OC = \tan^{-1} \cos a \sin B$; we ought to have $A = BO + OB$; that is, in

$$A = \tan^{-1} \cos a \sin B + \tan^{-1} \cos a \tan b,$$

which, observing that $\sin \theta \cdot \sin a = \sin b \cdot \sin B$, and $\sin \theta \cdot \sin A = \sin b \cdot \sin A = a$, that is equivalent to, the above formula.

Observe in particular that when A is at the centre, p is $= 0$, and the formula $\hat{A} = \cos \theta$ is, a, b , or say in an angle at the centre, $\hat{A} = 0$.

p. 412

I return to the expression for $\cosh \hat{a}$; in explanation of its meaning let the distances $\hat{a}, \hat{c}$ be θ, η respectively and let the angle BOC be θ ; to find θ we have only to take θ' as θ ; that is, in the formula for $\cosh \hat{a}$ to write $a = 0$, etc., find $\cosh \hat{a} = \frac{1 - \sin a \sin b \cos \theta}{\cos b \cos c}$; and similarly for $\cos \alpha$ and $\sin \alpha$, where $\alpha = OB'$, and $\sin \alpha = OB \sin C$; we have similarly $\sin a = OB \sin B$, viz. we have

$$\cos \hat{a} = \cos b \cos c \cdot \cosh \hat{a},$$

$$\sin \alpha = \cos a \sin b \cdot \cos C,$$

or, what is the same thing, θ is

$$\cos \hat{a} = \cos b \cos c + \sin b \sin c \cdot \cos C,$$

$$= \cosh \hat{b} \cdot \cosh \hat{c} + \sinh \hat{b} \sinh \hat{c} \cos C,$$

viz. as will presently appear, this is the formula for $\cos C$ in the triangle $\hat{B}\hat{O}\hat{C}$.

From the above formula

$$\cosh \hat{a} = \frac{1 - \sin a \sin b \cos \theta}{\cos b \cos c}.$$

and

$$\sinh \hat{a} = \frac{a \cdot p \sin A}{\cos b \cos c} = \frac{a \cdot p \sin A}{\cos b \cos c \cdot \sin \theta \sin \theta},$$

and the like formulae for $\hat{b}, \hat{c}, B, C$ in any way be shown that in the triangle ABC we have

$$\frac{\cos \hat{A} + \cos \hat{B} \cos \hat{C}}{\sin \hat{B} \sin \hat{C}} = \cos \hat{a},$$

In fact, substituting the foregoing values, this equation becomes

$$(1 - \sin a) \cos A + (1 - \sin a) \cos b \cos C (1 - \sin a \cos C \cos A) = 1 - \sin b \sin c \cos A \\ - \sin B \sin C (1 - \sin a \sin c) \cos A$$

that is,

$$\cos A + \cos B \cos C - \sin a \sin b \cos C - \sin a \cos b \cos C + \sin a \sin b \cos C + \sin a \cos c \cos C \sin a \\ - \sin B \sin C (1 - \cos a),$$

or, what is the same thing,

$$\sin^2 a (\cos B \cos C - \cos A) = \sin B \sin C + \sin a \sin b \cos A \\ + \sin a \sin b \cos a \cos a \cos C \sin a$$

that is,

$$(\sin a \cos B + \sin a \sin c \cos C)(B + \sin a \cos a) - (\sin a \cos a) \cos A = \sin a \sin b \cos A$$

a relation which I proceed to verify.

p. 413

We may, from the formula

$$\sin^{-1} \sin g = \sin^2 a = 1 - \sin a \sin b \cos b, \&c.$$

but, more simply, geometrically as presented above, deduce

$$\sin a \cos B + \sin a \cos C = \sin a (\cos a \sin a + \sin a \cos B + \sin a \cos C),$$

and thence

$$\sin a \cos A + \sin a \cos B + \sin a \cos C = \sin a,$$

$$\begin{aligned} (\sin a \cos B + \sin a \cos C)(\sin a \cos a + \sin a) &= \sin a (\sin a (\sin a + \cos a) \sin a + \cos a \sin a + \sin a) \\ &= a \cdot \sin C (1 - \sin g \cos a), \end{aligned}$$

which, observing that $\sin g \sin a \cdot \sin A = a \cdot p$ and consequently $a, b, A, B, \cos A \cdot \cos B = b \cdot c$, and so multiplying each side by $\sin a$, that is, multiplying the second side by $\cos A \cdot \cos B = a^2 \cdot p^2$ in the case of the surfaces in question, the formulae for $\hat{a}$ is

$$\sin a \sin a \cdot \sin a \cdot \cos A = a \cdot p \sin C (1 - \sin a \cos a)$$

and the other of the equations in question is proved in the same manner.

From the formula for each a , we find

$$\sinh^2 \hat{a} = \frac{1}{\sin a \cdot \cos a} \Lambda,$$

where

$$2\sqrt{-} = (1 - \cos a) \cdot a + \cos a \cdot b + \cos a \cdot c (1 - 3a \sin a \cdot b \cdot \cos a),$$

whence also

$$\sinh^2 \hat{a} : \sinh^2 \hat{b} : \sinh^2 \hat{c} = \sin a \cdot a : \sin a \cdot b : \sin a \cdot c$$

and we can also obtain

$$\cos \hat{a} = \frac{\cos a + \cos a \cos b \cos ac}{\sinh b \sinh c},$$

So that the formulae are in fact similar to those of spherical trigonometry with only $\cos \hat{a}, \sin \hat{a}, a$, instead of $\cos a, \sin a$, etc. The before-mentioned formula for each $\hat{a}$ in terms of a, b, c ; it is obviously a particular case of the last-mentioned formula for $\cos \hat{A}$.

Cambridge, 11 May, 1872.

[529]

A “Smith’s Prize” Paper(?); Solutions.

[From the Oxford, Cambridge and Dublin Messenger of Mathematics, vol. IV. (1868), pp. 201–220.]

p. 414

1. *Find the form of a function of n given number of letters, which has two and only two values.*

It is required to find the general form of a function $\phi(a, b, c, \dots, k)$, rational and integral, which for all permutations whatever of the letters has two and only two values.

Suppose that any particular permutation of the letters changes $\phi(a, b, c, \dots, k)$ into $\phi_1(a, b, c, \dots, k)$; then any permutation of the letters will either leave the functions ϕ, ϕ_1 , each of them unaltered, or it will change ϕ into ϕ_1 and ϕ_1 into ϕ . Hence $\phi - \phi_1$ is a symmetrical function of the letters, say

$$\phi + \phi_1 = 2L.$$

$\phi - \phi_1$ is a function which by a permutation of the letters is either unaltered, or simply changes its sign; and it is to be shown that, writing for shortness V to denote the product $(a - b)(a - c) \dots (b - c) \dots (k - 1)$, of the differences of the letters, and denoting by SV a symmetrical function of $a, b, c, \dots, k$, the form of $\phi - \phi_1$ is

$$\phi - \phi_1 = 2VM.$$

These equations give

$$\phi = L + VM,$$

which is the general theorem required; viz. the function ϕ has thus only the two values $L + VM$ and $L - VM$, where L, M are symmetrical functions.

To prove the subsidiary theorem, observe that there is at least one interchange of two letters which changes ϕ into ϕ_1 , or otherwise $\phi = \phi_1$, would be a symmetrical function.

* Set by me, for the Master of Trinity, Thursday, January 30, 1868.

p. 415

let this be (a, b) . Then $\phi - \phi_1$, changing its sign for the interchange in question, must vanish for $a = b$; that is, the product $\phi - \phi_1$ must contain $(a - b)$ as a factor: it is in the power $(a - b)^{2k+1}$; the product $\phi - \phi_1$, vanishing for $a = b$, vanishes for $a = b$, that is, contains $(a - b)$ as a factor: similarly $(a - c)$, $(a - d) \dots (b - c) \dots$ are factors, but not every other letter; and consequently $\phi - \phi_1$ contains V as a factor; we have therefore $\phi - \phi_1 = VQ$.

Hence the interchange of any two letters a, b , say of the letters (a, b) , changes Q into $-Q$; that is, in the new equation, VQ becomes $-VQ$, viz. we have VQ . $AV = -VQ$, suppose in general, that lines any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then, the interchange of (b, c) , $(c, d), \dots$ is to leave Q unaltered, that is, Q is unaltered by the interchange of any two adjacent letters, $a = b$, $b = c$, etc.; but every interchange may be obtained by means of the interchange of two adjacent letters; that is, every interchange leaves Q unaltered; that is, Q is a symmetrical function $= SM$, and thence $\phi - \phi_1$, which is the subsidiary theorem. But we have VM , $\phi = L + VM$, which is the form of the function ϕ .

Observing that if a quadratic function $ax^2 + 2bxy + cy^2$ has two equal roots, then, since the radical sign disappears, the function ϕ has only one value $= L$, and from two letters which changes ϕ to ϕ_1 , that of V being a symmetrical function, is odd or even.

2. Express $\frac{cx+d}{ax+b}$ in the form of two cubic factors; and show generally that, for any prime number p , the function $\frac{cx^{p-1}+dx+\dots}{ax^{p-1}+bx^{p-2}+\dots+c}$ may be broken up into two factors of the order $\frac{1}{2}(p-1)$ by means of a quadratic radical.

Let r be any root of the given equation, then

$$x^{p-1} + x^{p-2} + \dots + 1 = 0,$$

the roots of this equation are $r, r^2, \dots, r^{p-1}$, where r is an imaginary p th root of unity.

Hence

$$x^{p-1} + x^{p-2} + \dots + 1 = (x - r)(x - r^2) \dots (x - r^{p-1}),$$

and denoting the two cubic factors by ρ_1, ρ_2 we have

$$\begin{aligned} \rho_1 &= (x - r)(x - r^2)(x - r^4) \dots, \\ \rho_2 &= (x - r^3)(x - r^5)(x - r^6) \dots, \end{aligned}$$

we have

$$\rho_1 + \rho_2 = x^{p-1} + x^{p-2} + \dots + 1,$$

and thence

$$\rho_1 \rho_2 = x^{2p-2} + \dots (1 + x + r^2 + \dots + x^{2(p-2)+1}).$$

$$\rho_1 = x^{p-1} + \dots + r^p,$$

$$\rho_1 + \rho_2 = x^{p-1} + \dots (1 + x + r^2 + \dots + x^{2(p-2)+1});$$

Hence ρ_1, ρ_2 are the roots of the quadratic equation

$$y^2 - (3p^2 - 2x \pm 1)y - 5a^2 + \dots + (4 - 2)p^2 + (2 - 1) = 0.$$

p. 416

Solving the equation, observing that the quantity under the radical sign must of necessity be a square, and that its value is

$$(2x^2 + x^{p-1} + \dots + 2)^2 - 4(x^{p-1} + x^{p-2} + \dots + x^2 + x + 1),$$

which is in fact

$$= -7(x^2 + 1)^2,$$

so that the roots y_1, y_2 , that is, the required cubic factors, are

$$\frac{1}{2}(2x^2 + x^{p-1} - \dots + 2) \pm \frac{1}{2}(x^2 + 1)\sqrt{-7}.$$

Generally, for any prime exponent p , denoting by r any root of $x^{p-1} + \dots + 1 = 0$, and writing

$$y = \rho_1 = (x - r)(x - r^2) \dots = e^{p-1} + \dots + b,$$

where $a, b, \dots$ are the quadratic residues, and $a, b, \dots$ the quadratic non-residues of p , we find

$$y' + y' = (y + y)^2,$$

where P is a function of x of the order $\frac{1}{2}(p-1)$. Putting also $y'y' = (P - 4Q)$, where Q is a function of x of the order $\frac{1}{2}(p-1)$, then y_1, y_2 are the values of a quadratic equation: $y' - Py + Q = 0$, and these roots are

$$\frac{1}{2}(P \pm \sqrt{(P^2 - 4Q)}).$$

Here, as is known, $P^2 - 4Q$, in a perfect square, $= qP^2$, E a rational function of the two coefficients, and the quadratic radical is then $\sqrt{qP^2}$, that is, the radicals to be employed of the above-mentioned function of the coefficients can be obtained by proper determinations of the above-mentioned function T'^{-1} in the symmetrical function which I call

$$\frac{1}{2}(P \pm \sqrt{qP}).$$

2. Is a Map of the World, wherein the meridians are projected into right lines meeting in a point, the inclination of any two of the lines being equal to that of the two meridians, and the parallels into circles about the point as centre, the radius of the circle being a given function of the latitude; compare on the plane scale of equal parts at any two points, calculated by means of a proper determination of the above-mentioned function of the latitude in question.

Let the longitude and colatitude be

on the sphere: l, λ ,

in the map: l', r ,

then the projection is such, that $l' = l, r = f(\lambda)$, a given function of λ .

The lengths of corresponding linear elements in the direction of a meridian, and perpendicular to it, are

$$de, \sin e dl,$$

$$dr', r' dl';$$